

Assessing the Landscape of Data for Predicting TCRs from Tumour Mutations

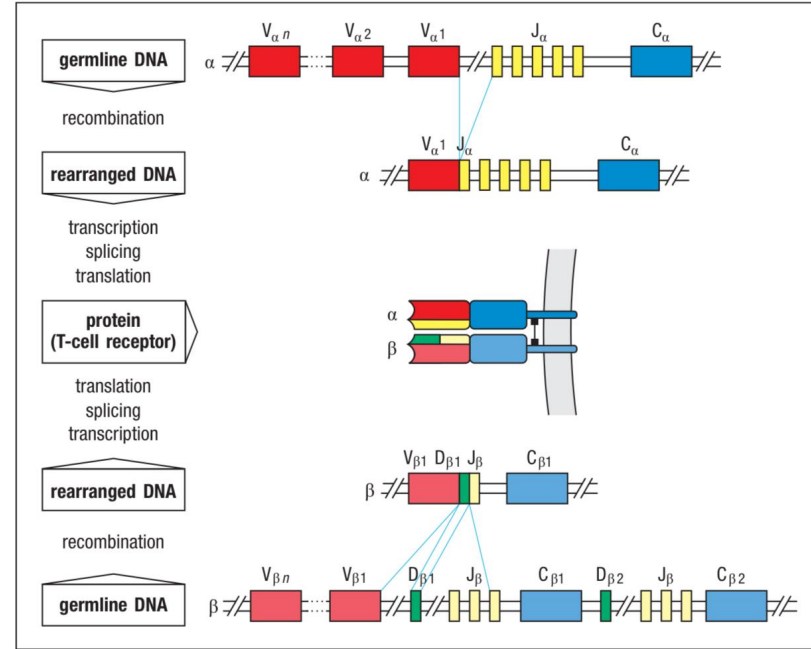
April 14, 2025
Victoria Pergola
Campbell Lab

Overview

1. TCR Background
2. Motivation
3. Methods
4. TCR Data Summary
5. Tumour Mutation Data Summary
6. Logistic Regression Results
7. Next Steps
8. Conclusion

TCR Structure

- T-cells are crucial in adaptive immunity, recognizing antigens via the T-cell receptor (TCR).
- The TCR is formed by V(D)J recombination. The CDR3 region, formed by joining variable (V), diversity (D), and joining (J) gene segments, is the most diverse and binds antigens.

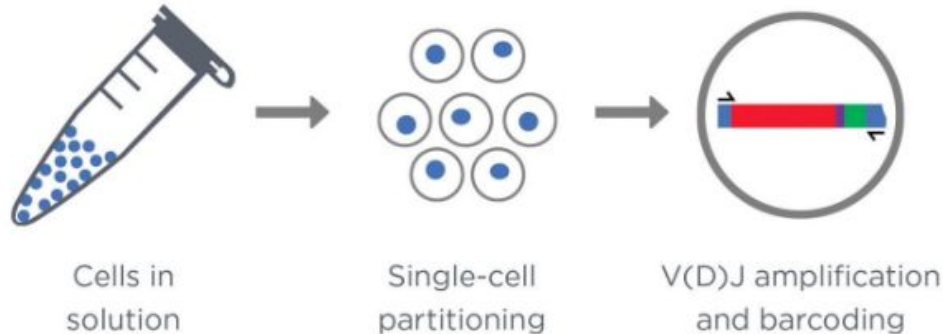


TCR Sequencing

- Uses PCR amplification of V, D, J and some C segments from RNA or DNA to capture TCR sequences from single cell library
- Since these segments are rare this helps us get reads

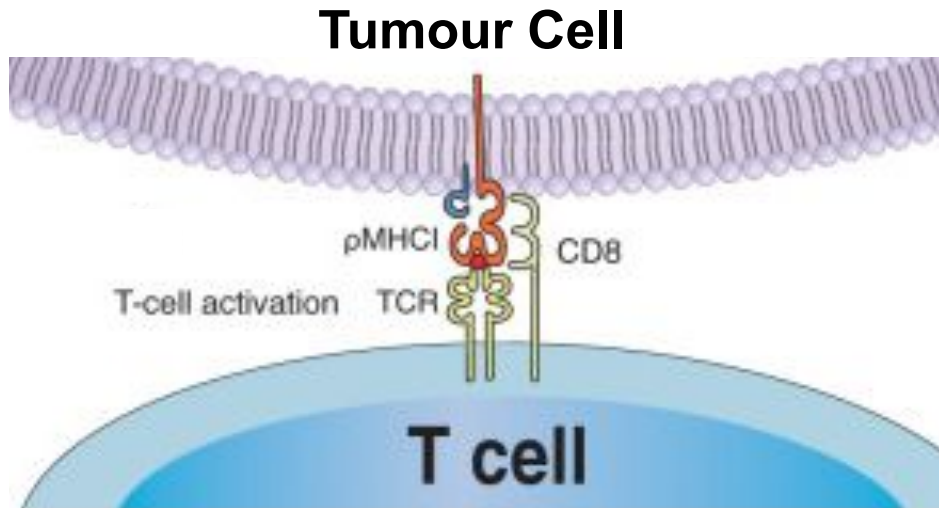
Single-Cell Immuno-Profiling

For physical pairing of chains



T-cell Recognition of Tumours

- Tumour cells can be recognized by T-cells if they present neoantigens
 - Peptides derived from somatic mutations



Scientific Question:

Can co-occurrence between tumour mutations and TCR sequences be leveraged to predict the presence of tumour neoantigens using computational models?

Co-occurrence Between Somatic Mutations and TCR CDR3 Motifs

Panel A: Mutation–CDR3 Co-occurrence

- Identified mutation–TCR motif pairs that co-occur more than expected by chance
- Used permutation testing with $\text{FDR} \leq 0.05$ to assess significance
- Suggest specific TCR motifs may target recurrent mutations

Panel B: MHC Binding Predictions

- Predicted binding of mutant peptides to MHC class I using NetMHC4.0
- Mutated peptides often showed stronger binding than wild-type versions
- Supports idea that these mutations may produce neoantigens

From bulk RNA-seq data.

9,142 tumour samples, 29 cancer types

600,000 CDR3 sequences, 15% were full-length

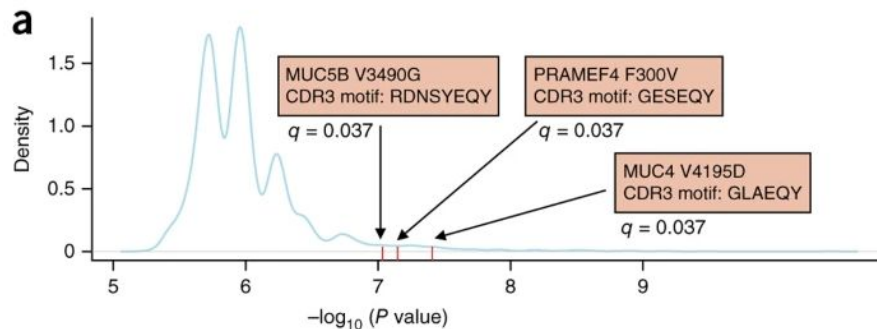
Analysis | Published: 30 May 2016

Landscape of tumor-infiltrating T cell repertoire of human cancers

Bo Li, Taiwen Li, Jean-Christophe Pignon, Binbin Wang, Jinzeng Wang, Sachet A Shukla, Ruoxu Dou, Qianming Chen, F Stephen Hodi, Toni K Choueiri, Catherine Wu, Nir Hacohen, Sabina Signoretti, Jun S Liu & X Shirley Liu

Nature Genetics 48, 725–732 (2016) | [Cite this article](#)

27k Accesses | 226 Citations | 69 Altmetric | [Metrics](#)



b

Gene	Peptide	MHC I allele	Rank	Predicted binding
PRAMEF4	CLKTSLKVL	B*08:01	0.4%	Strong
PRAMEF4	KTSLKVLTI	B*58:01	0.4%	Strong
PRAMEF4	KVLTITNCV	A*02:01	1%	Weak
PRAMEF4	KVLTITNCV	A*30:01	0.7%	Weak

Current Landscape of Neoantigen Prediction

Alspach et al. 2019

- Trained on known **MHC-binding** peptides to predict neoantigens.

Lu et al. 2021

- Learn binding specificity from TCR and pMHC **sequence data**.

Jeong et al. 2024

- Used NetMHCpan-4.1 with tumour mutation + mass spectrometry ligand data to predict **MHC-binding** neoantigen candidates

Article | Published: 23 October 2019

MHC-II neoantigens shape tumour immunity and response to immunotherapy

[Elise Alspach](#), [Danielle M. Lussier](#), [Alexander P. Miceli](#), [Ilya Kizhvatov](#), [Michel DuPage](#), [Adrienne M. Luoma](#), [Wei Meng](#), [Cheryl F. Lichti](#), [Ekaterina Esaulova](#), [Anthony N. Vomund](#), [Daniele Runci](#), [Jeffrey P. Ward](#), [Matthew M. Gubin](#), [Ruan F. V. Medrano](#), [Cora D. Arthur](#), [J. Michael White](#), [Kathleen C. F. Sheehan](#), [Alex Chen](#), [Kai W. Wucherpfennig](#), [Tyler Jacks](#), [Emil R. Unanue](#), [Maxim N. Artyomov](#) & [Robert D. Schreiber](#) 

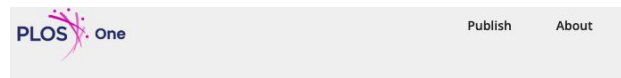
[Nature](#) 574, 696–701 (2019) | [Cite this article](#)

Article | Published: 23 September 2021

Deep learning-based prediction of the T cell receptor–antigen binding specificity

[Tianshi Lu](#), [Ze Zhang](#), [James Zhu](#), [Yunguan Wang](#), [Peixin Jiang](#), [Xue Xiao](#), [Chantale Bernatchez](#), [John V. Heymach](#), [Don L. Gibbons](#), [Jun Wang](#), [Lin Xu](#), [Alexandre Reuben](#)  & [Tao Wang](#) 

[Nature Machine Intelligence](#) 3, 864–875 (2021) | [Cite this article](#)



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RESEARCH ARTICLE

GraphMHC: Neoantigen prediction model applying the graph neural network to molecular structure

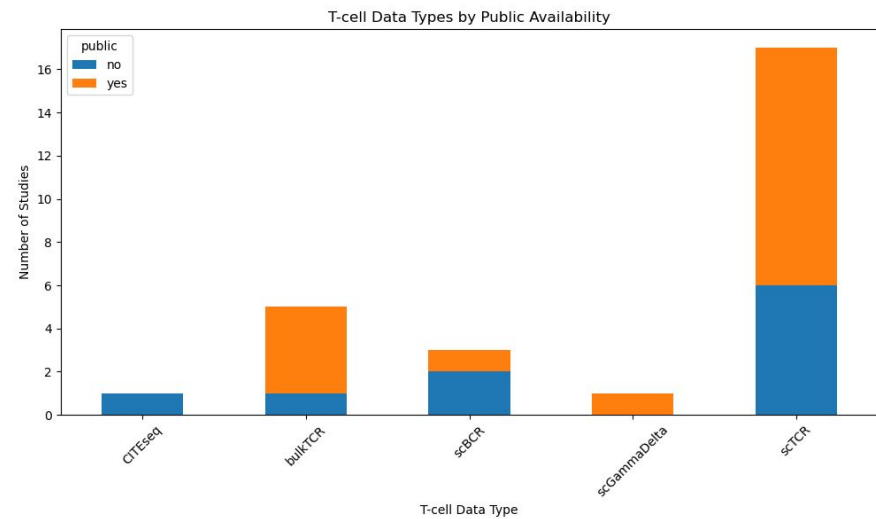
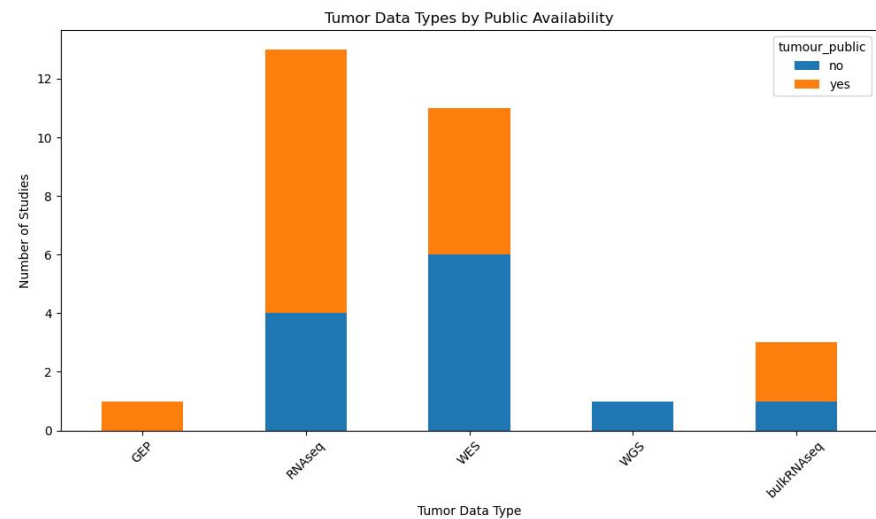
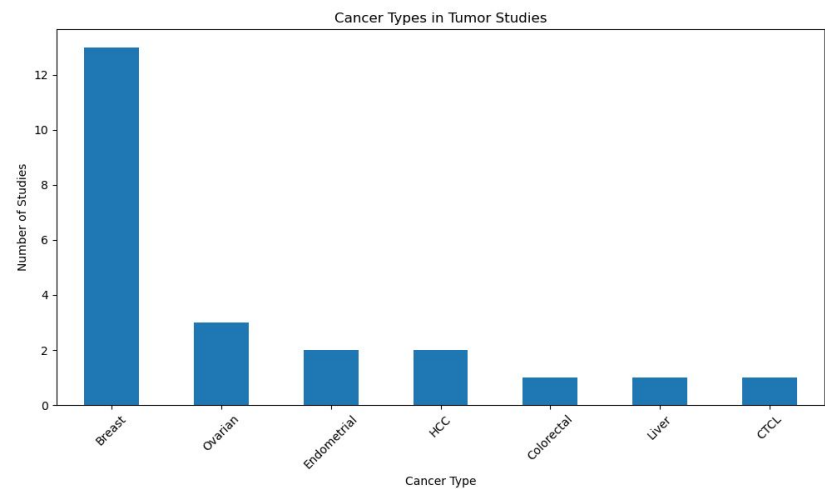
[Hoyeon Jeong](#), [Young-Rae Cho](#), [Jungsoo Gim](#), [Seung-Kuy Cha](#), [Maengsup Kim](#), [Dae Ryong Kang](#) 

Published: March 27, 2024 • <https://doi.org/10.1371/journal.pone.0291223>

Motivation and Goals

1. Co-occurrence relationship between TCRs and tumour mutations is currently underexplored in literature
2. This project aims to assess the feasibility of co occurrence-based prediction by analyzing existing datasets

Literature Review



Matched Mutation and TCR CTCL Dataset

- Investigated transformed CTCL (tCTCL), aggressive form of CTCL
 - Rare type of non-Hodgkin lymphoma characterized by malignant T-cells that primarily infiltrate the skin.
- Contained 8 matched WES and VDJseq patient samples
- Publicly available**

RESEARCH ARTICLES | MAY 02 2022

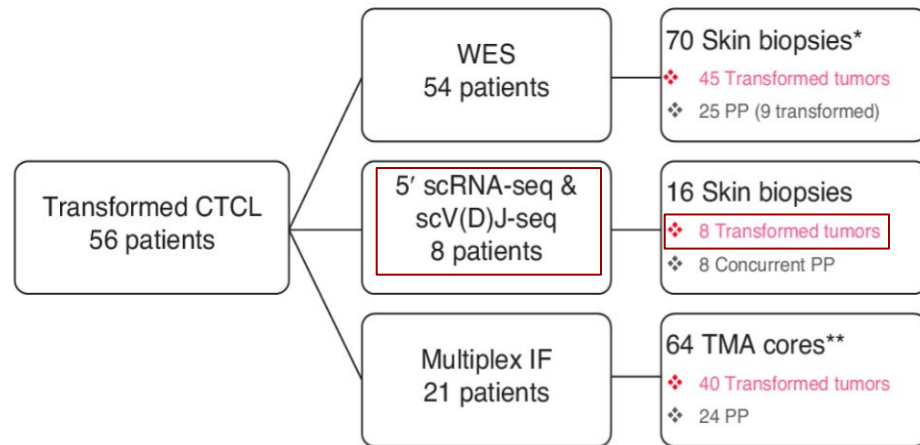
Genomic and Single-Cell Landscape Reveals Novel Drivers and Therapeutic Vulnerabilities of Transformed Cutaneous T-cell Lymphoma **FREE**

Xiaofei Song; Shilun Chang; Lucia Seminario-Vidal; Alvaro de Mingo Pulido; Leticia Tordesillas; Xingzhi Song; Rhianna A. Reed; Andrea Harkins; Shannen Whiddon; Jonathan V. Nguyen; Carlos Moran Segura; Chaomei Zhang; Sean Yoder; Zena Sayegh; Yun Zhao; Jane L. Messina; Carly M. Harro; Xiaohui Zhang; José R. Conejo-García; Anders Berglund; Lubomir Sokol; Jianhua Zhang; Paulo C. Rodriguez; James J. Mulé; Andrew P. Futreal; Kenneth Y. Tsai; Pei-Ling Chen



+ Author & Article Information

Cancer Discov (2022) 12 (5): 1294–1313.



Methods

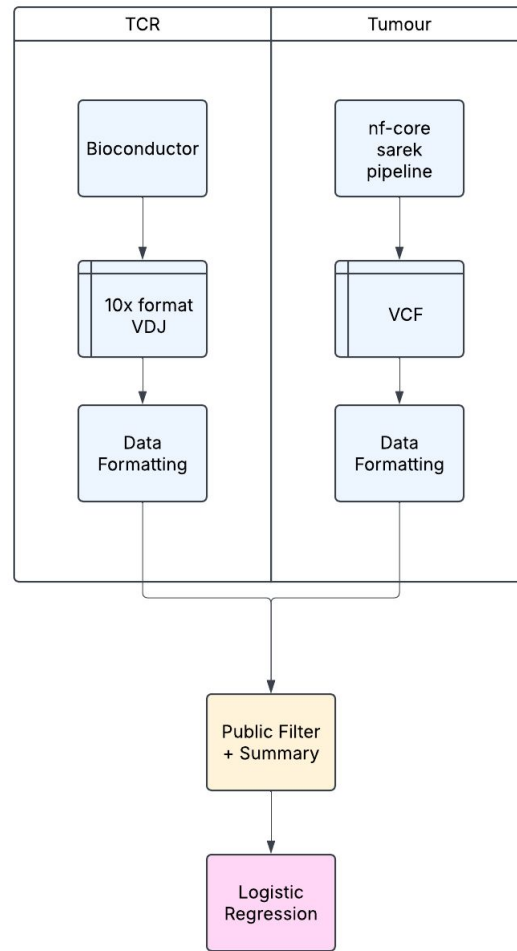
1. Mutation calling via Sarek + DeepVariant

- Filtered using DeepVariant filter criteria → shown to outperform other variant calling tools
- Analyzed using maftools

2. TCRs derived from single-cell 10x data

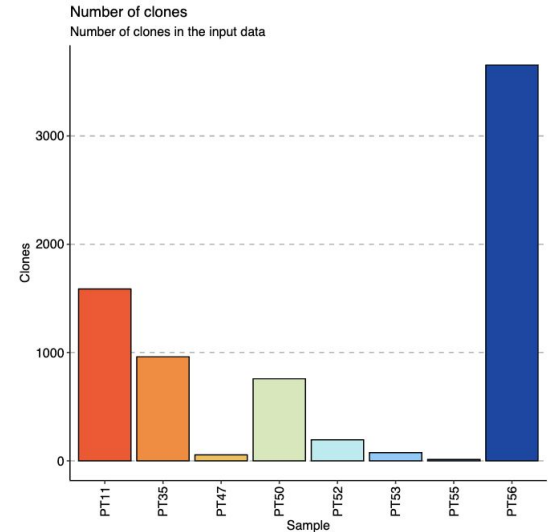
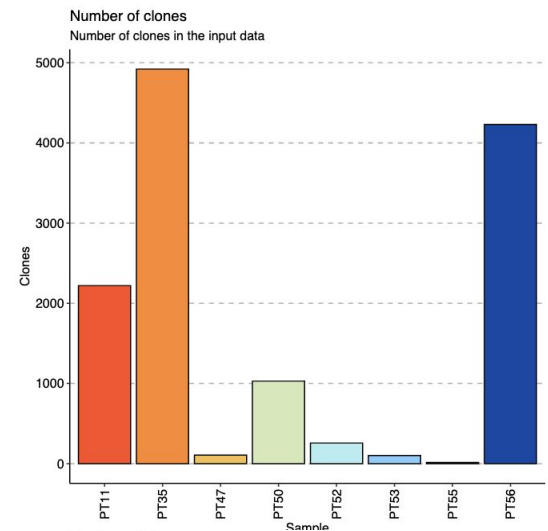
- Filtered out barcodes with multiple α or β reads or low confidence
- Analyzed using immunarch

3. Built binary logistic regression models in scikit-learn using public TCRs



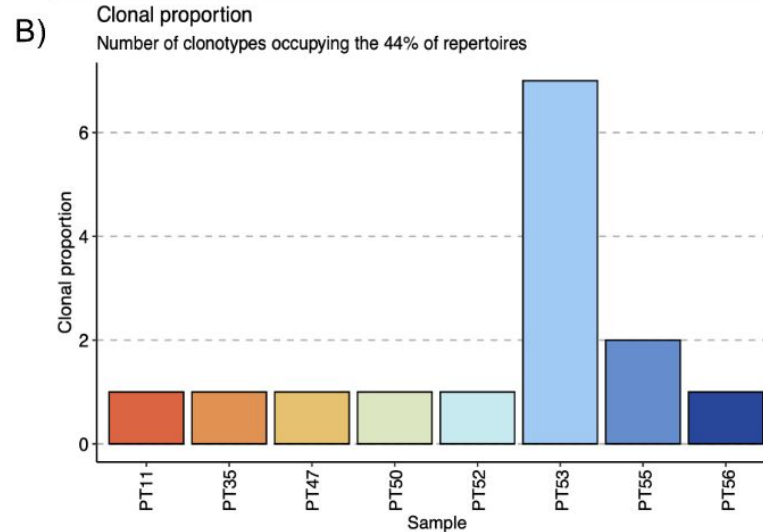
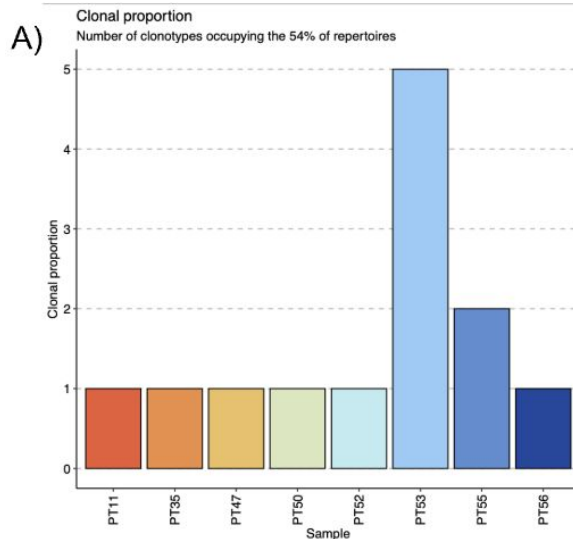
Total T-cell Counts

- PT56 had the highest T-cell count after filtering
- Filtering ensured there were not multiple reads for α or β chains.
- Low counts in samples such as PT55 may be due to poor quality of data uploaded by authors



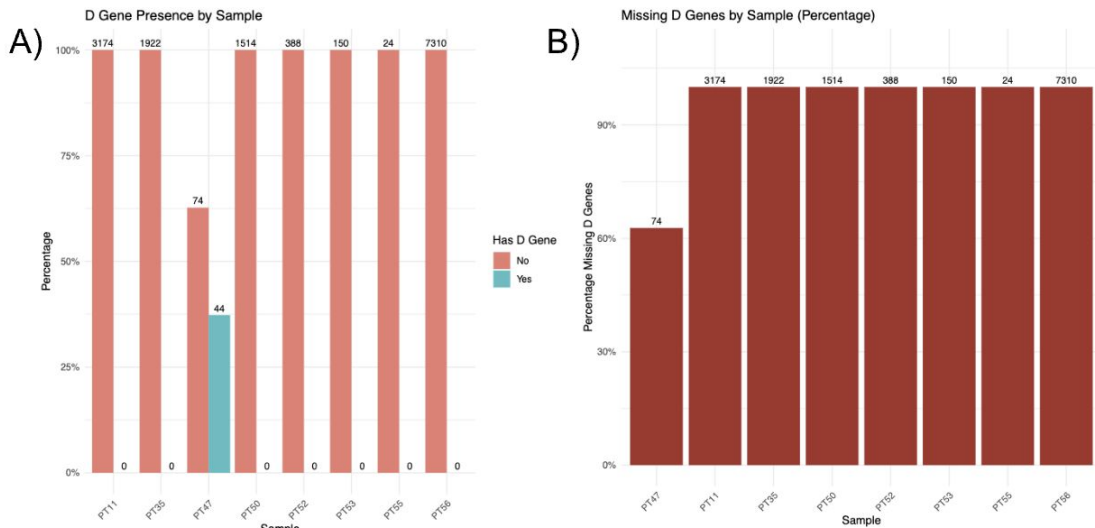
Clonal Proportion

- Dominance of one cell is likely the malignant T-cell
 - Potential limitation with using a CTCL dataset.
- Indicates hyperexpansion of one T cell clone within tumour



D Gene Presence

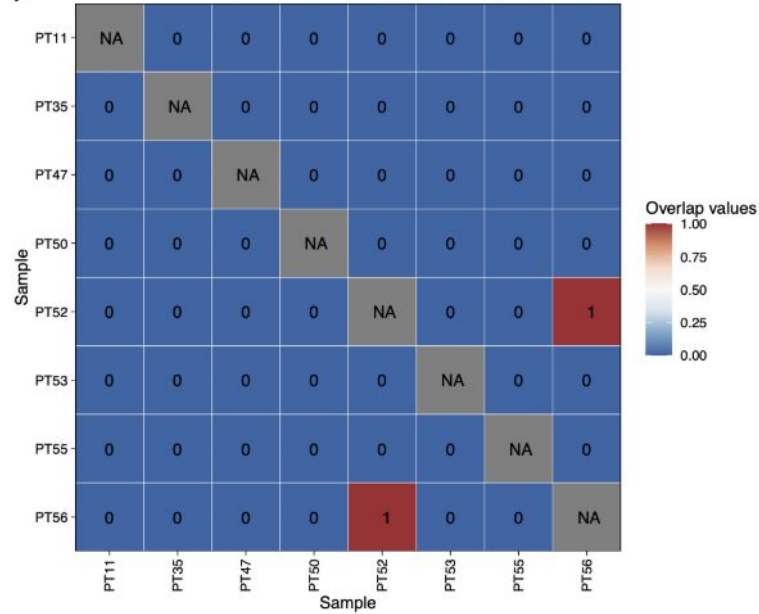
- In single-cell VDJ sequencing the D region is often not confidently assigned.
 - Only 2 annotated D genes in the 10x reference genome
- Defined public TCRs using only V and J gene combinations
 - improves comparability but may reduce sensitivity in identifying truly public TCRs.



Repertoire Overlap

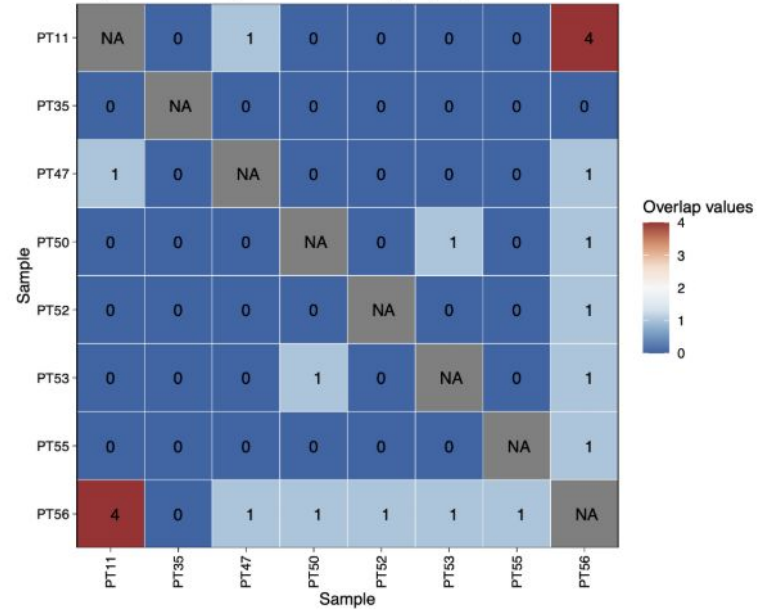
A)

Repertoire Overlap of Public Clonotypes (aa+v+j)



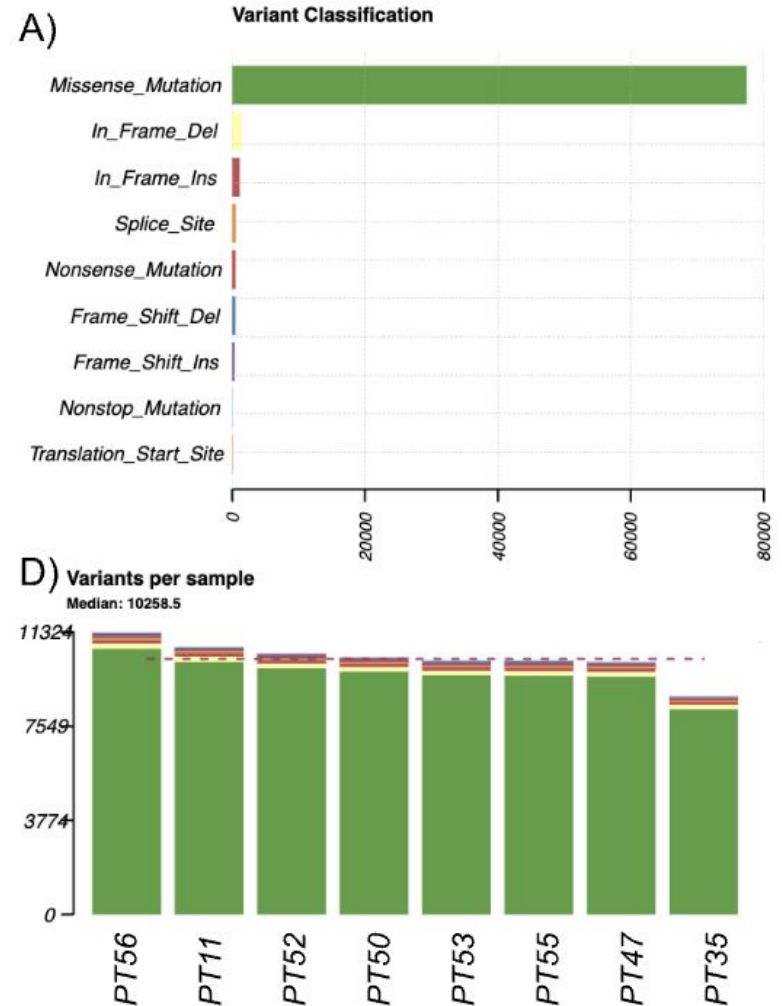
B)

Repertoire Overlap of Public Clonotypes (v+j)



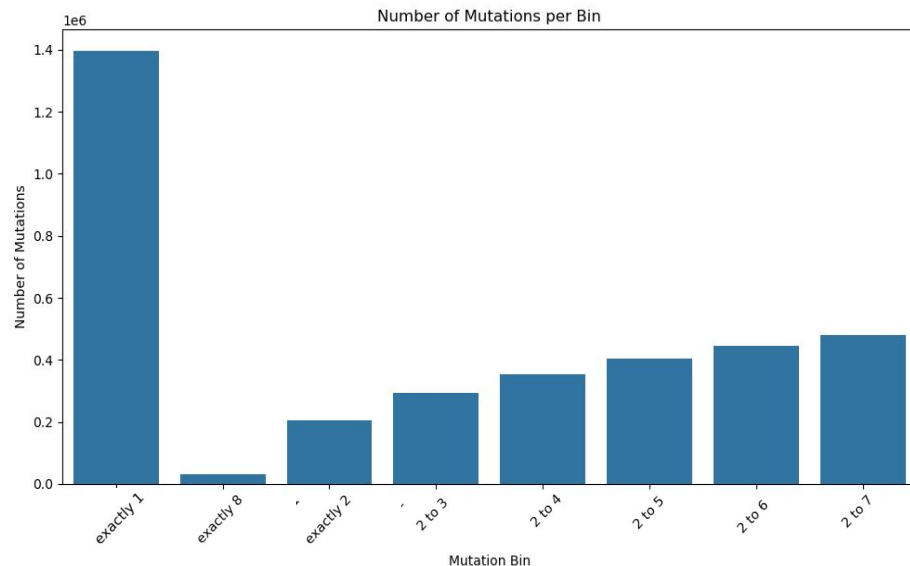
Tumour Mutation Overview

- Majority of mutations are missense
- Mutation counts vary:
PT56 had ~11,000, PT35 ~8000



Binary Logistic Regression

- Built models to predict presence of public V+J TCRs
- Mutations grouped into bins by prevalence
 - Compare performance across bins



Key Takeaways

- WES is rare and hard to access—dataset size is a barrier.
 - Matched WES and TCRseq is rare
 - WES is the most restricted data type
 - Using WES for the project was a major barrier to collecting a large dataset
- D region is not annotated in most single-cell datasets → limits clonotype resolution.
- A larger dataset is needed in order to assess accuracy with a reasonable train-test split

Next Steps

- **Expand dataset:** More patients, focused tumour type, ideally with bulkRNAseq
 - Get more reads, easier to infer mutations as there are more established tools
- **Proper validation:** Use test/train splits to assess generalizability of the model.
- **Generalizability:** Test across cancers, look for convergent TCR responses.

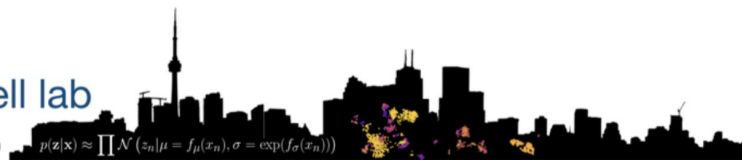
Acknowledgements

- Sarah Asbury
- Shanza Ayub
- Chengxin Yu
- Hassaan Maan
- Michael Geuenich
- Matthew Watson
- Haifen Chen
- Alina Selega
- Jett Lee
- Cait Harrigan
- Tan Tiak Ju
- Vesal Kasmaeifar
- Morris Greenberg
- All members of the Campbell Lab

Supervisor

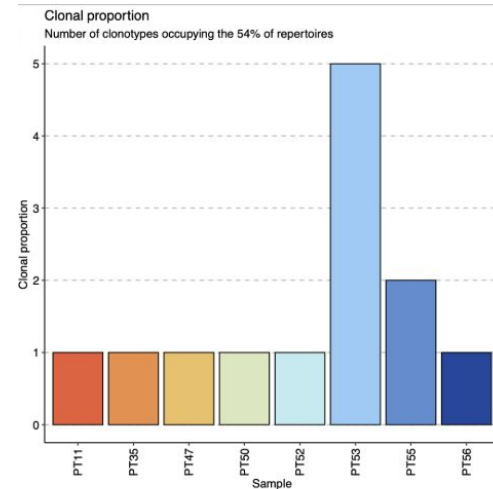
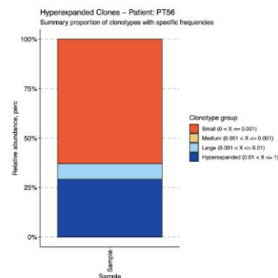
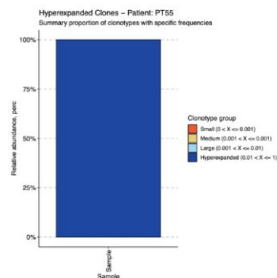
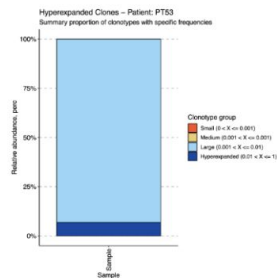
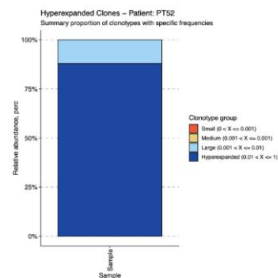
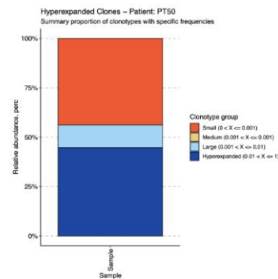
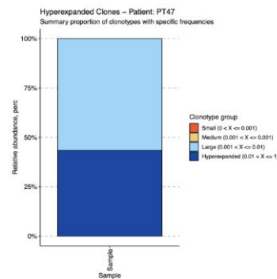
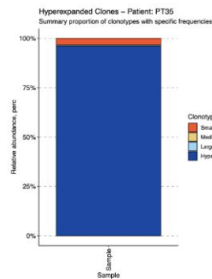
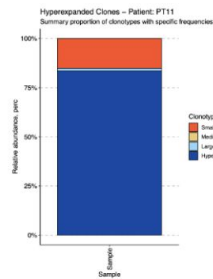
Dr. Kieran Campbell

Campbell lab
Toronto



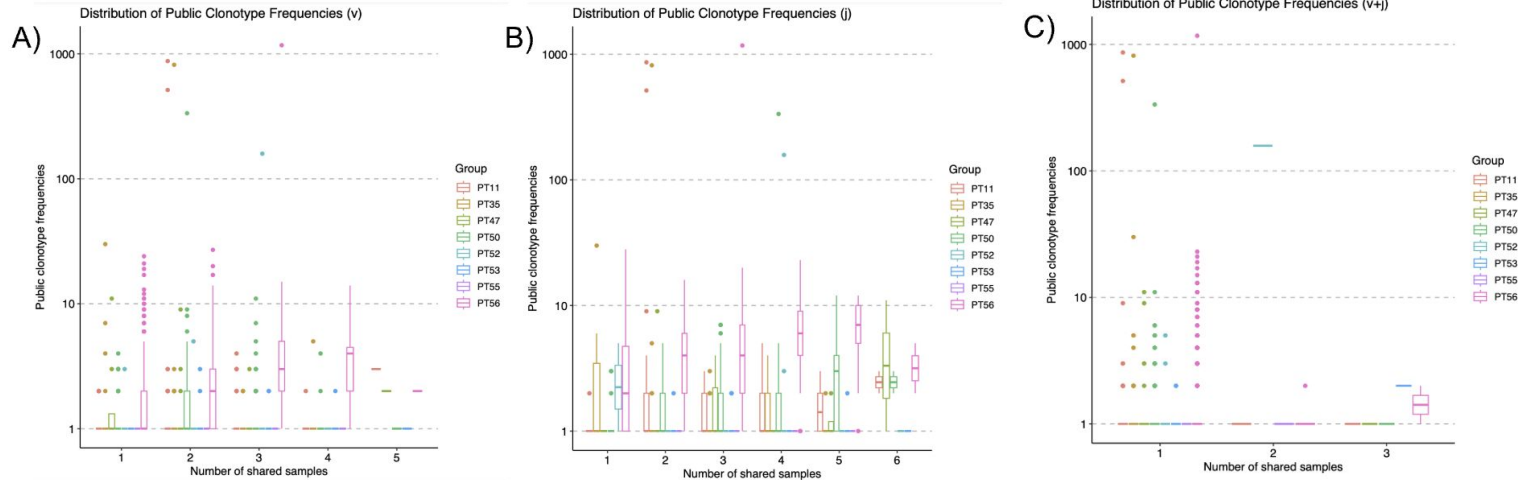
Supplementary Slides

Clonal Proportion cont.

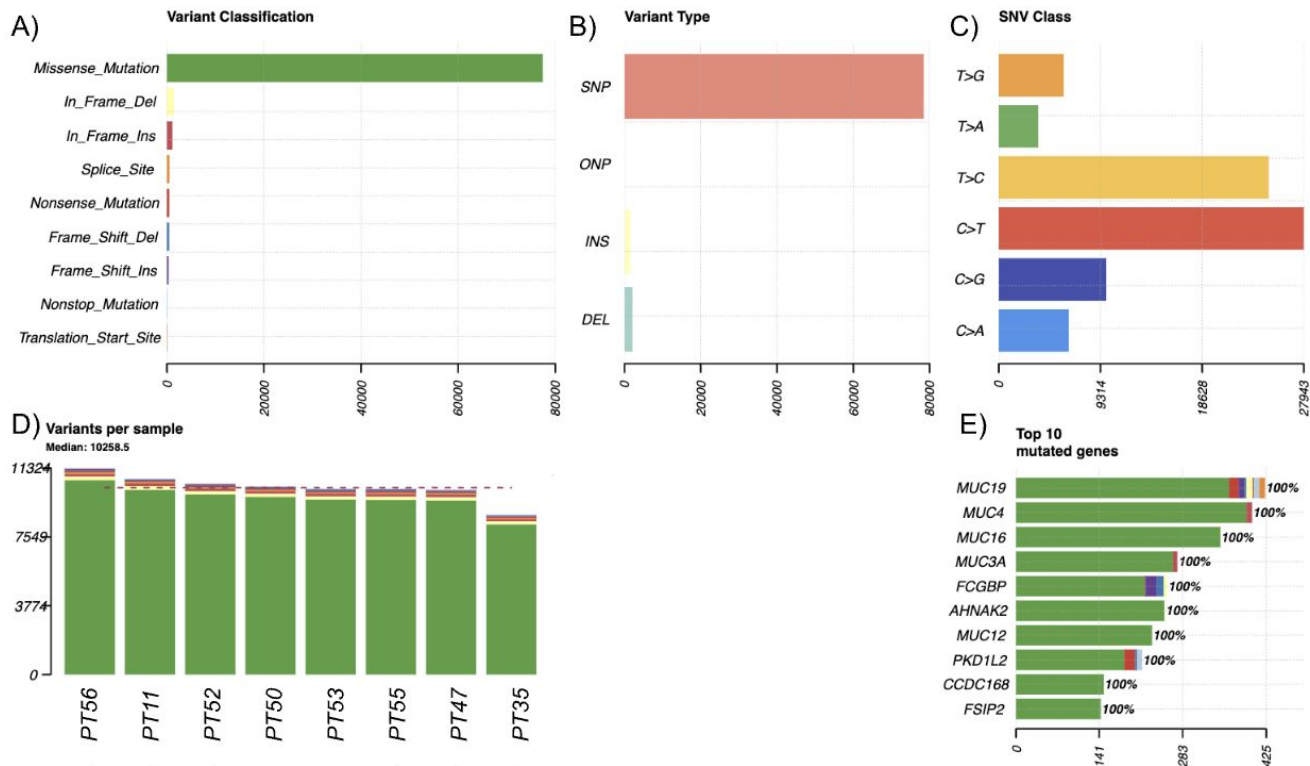


Distribution of Public Clonotype Frequencies

- Most high-frequency V+J clonotypes are only found in one sample. (private)
- Public clonotypes are usually more rare within a sample
- Public V or J alone appear in more samples than V+J



Tumour Mutation Overview



Montemurro, 2021

Fig. 3: Performance of models trained on paired-chain data.

